[INCH-POUND]
A-A-59292
21 July 1998
SUPERSEDING
MIL-W-81319A(AS)
22 August 1967

COMMERCIAL ITEM DESCRIPTION

ABRASIVE WHEELS, NON-WOVEN NYLON, RESIN REINFORCED

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers reinforced abrasive wheels consisting of aluminum oxide (Al_2O_3) abrasive particles, non-woven nylon fibers, and a thermosettting resin binder. The wheels are for use in removing contaminants, oxides, and other corrosion products from various surfaces. This CID only includes the abrasive wheels from MIL-W-81319 that have been direct purchased by the Government.

2. CLASSIFICATION. The abrasive wheels shall be of the following classes and grades:

2.1 Classes.

Class 1 - Convolutely wound
Class 2 - Laminated

2.2 Grades.

Grade B - Fine grit size (available for all classes)
Grade C - Medium grit size

3. SALIENT CHARACTERISTICS.

3.1 Materials. The abrasive wheels shall consist of sized abrasive particles, non-woven nylon fibers and a good quality thermosetting resin binder.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this documents should be sent to: Commander, Naval Air Warfare Center Aircraft Division, Code 41400B120-3, Highway 547, Lakehurst, NJ 08733-5100.

FSC 5345

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3.2 Abrasive. The abrasive shall be 94% minimum Al_2O_3, 9.4 MOH's scale, and shall be suitable for fine finish plus stock removal.

3.2.1 Abrasive Grit Size. The abrasive particle grit size shall be as specified in Table I. Grit size grading shall be accomplished in accordance with ANSI B74.18 and shall be determined on particles that have been removed from a wheel.

TABLE I. Grit size.

GRADES	GRIT SIZE, microns	
	Minimum	Maximum
B (Fine)	180	240
C (Medium)	100	150

3.3 Non-Woven Nylon Fibers. The fiber used in the abrasive wheel shall be nylon 6 or 6-6, in accordance with Table II.

TABLE II. Nylon fiber properties.

PROPERTY	GRADE B	GRADE C
Denier	15 ±5	58 ±10
Crimps per inch	12 ±5	12 ±5
Heat set	Yes	Yes
Staple length, inch	1.5 ±0.125	2 ±0.188
Tenacity, grams/denier	3.0 to 6.0	2.5 to 4.5
% elongation	40 to 140	50 to 150

3.4 Wheel Dimensions. Abrasive wheel dimensions, tolerances, and the maximum safe revolutions per minute (RPM) in use speed shall be in accordance with ANSI B7.1 and Table III.

4. REGULATORY REQUIREMENTS

4.1 Regulatory Requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in inch-pound units, a request should be made to the contracting officer to determine if the product is acceptable.

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TABLE III. Wheel dimensions, tolerances, and safe in use speed.

CLASS	OD, inch 1/	ID, inch 2/	WIDTH, inch 3/ 4/	SAFE SPEED, RPM
1	20	12	1 to 7	1100
1	17	10	1 to 7	1500
1	16	10	1 to 7	1500
1	14	8	1 to 7	2000
1	12	5	1 to 7	2500
1	8	3	1 to 7	3200
1	6	1	1 to 7	4500
1	4	1	1 to 7	4500
2	6	1	0.25, 0.5, 0.75, 1	4800
2	3	0.5	0.25, 0.5, 0.75, 1	8000
2	2	0.25	0.25, 0.5	11000
2	1.5	0.125	0.25	13000

1/ OD tolerance up to and including 16 inches = ±0.188 inch.
Over 16 inches = ±0.125 inch.
2/ ID tolerance = ±0.010 inch.
3/ Width tolerance up to and including 1 inch = +0.188, –0 inch.
Over 1 inch = ±0.188 inch.

5. QUALITY ASSURANCE PROVISIONS.

5.1 Product Conformance. The product provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Market Acceptability. The following market acceptability criteria are necessary to document the quality of the product to be provided under this CID.

5.2.1 The company producing the abrasive wheels must have been producing a product meeting the requirements of this CID for at least one year.

6. PACKAGING

6.1 Preservation and Packing. Preservation and packing shall be as specified in the contract or order.

6.2 Marking. The wheels shall be identified and marked in accordance with ANSI B74.13.

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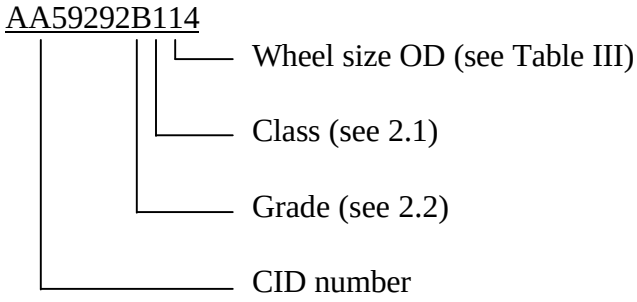
7. NOTES.

7.1 Part Identification Number (PIN). Wheels conforming to this CID shall be identified by a part number configuration consisting of the CID number, grade, class, and wheel size. An example of the part number configuration is as follows:



Part number AA59292B114 is for a Grade B, Class 1, 14-inch OD wheel.

7.2 National Stock Numbers.

5345-00-617-32165	5345-00-732-9977
5345-00-732-9978	5345-00-732-9988
5345-00-732-9989	5345-00-762-3166
5345-00-762-3175	5345-01-050-9703

7.2 Source of Documents.

7.2.1 ANSI Standards are available from the American National Standards Institute, Inc., 11 West 42nd Street, New York, NY 10036.

7.2 Ordering Information. The purchaser should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this commercial item description.
b. Grade, class, and wheel size.
c. Preservation, packing, and marking.

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MILITARY INTERESTS:

Custodians:
Army - MR
Navy - AS
Air Force - 99

Review Activity:
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITY:
GSA/FSS - 7FXE

Preparing Activity:
Navy - AS

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